Data. Alex et al. (2021) created the dataset by curating from 11 real-world datasets. The train-dev-test splits for the dataset are 550 training and 28712 development examples. Since the true test set is hidden, we report results on the development split.[92]

Pre-processing. To process the dataset, we extract the task-specific instructions from the original examples provided by Alex et al. (2021) at `https://github.com/oughtinc/raft-baselines/tree/master/example_prompts`.

Data Resources. We access the dataset at `https://huggingface.co/datasets/ought/raft`. The dataset is made available through our benchmark. The dataset has a datasheet available at `https://huggingface.co/datasets/ought/raft`.

C General metrics

C.1 Accuracy

By accuracy, we refer to the average correctness of models across all evaluation instances. Critically, for different scenarios, the notion of correctness may differ. Consequently, we enumerate the primary accuracy metrics we consider in this work, the scenarios for which they serve as the main accuracy metric, and the formal definition where relevant.

C.1.1 General

Exact match. The correctness condition for exact match is that the model generation match the correct reference exactly as strings. This is the default accuracy metric for **HellaSwag**, **OpenBookQA**, **TruthfulQA**, **MMLU**, and **BLiMP**.

Quasi-exact match. The correctness condition for quasi-exact match expands the exact match condition to identical strings up to some slightly post-processing of the model generation (i.e. lower-casing, removing whitespace and punctuation and articles). This is the default accuracy metric for **BoolQ**, **IMDB**, **CivilComments**, **RAFT**, **WikiFact**, synthetic reasoning (abstract symbols), **bAbI LegalSupport**, **LSAT**, **EntityMatching**, **DataImputation**.

F1. In contrast to the exact match methods, the correctness condition for F1 is more graded/not all-or-nothing to tolerate partial string overlap (see Rajpurkar et al., 2016). This is the default accuracy metric for **NaturalQuestions** (open-book), **NaturalQuestions** (closed-book), **NarrativeQA**, and **QuAC**.

C.1.2 Information Retrieval

RR@K. The correctness condition for RR@K depends on the reciprocal rank of the first relevant document, and is the default accuracy metric for **MS MARCO (regular)** for K=10.

Given $rank \in \{1, 2, 3, ...\}$ is the position of the *first* relevant document:

$$RR@K = \begin{cases} 1/rank, & \text{if } rank \leq K. \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

NDCG@K. The correctness condition for Normalized Discounted Cumulative Gain (NDCG; Järvelin & Kekäläinen, 2002) depends on the quality the quality of a set of graded rankings (i.e. assessing more than just binary relevance), and is the default accuracy metric for **MS MARCO (TREC)** for K=10.

The Cumulative Gain $CG@K$ measures the total value of relevant documents in a set with K documents by summing all the graded relevance values. To favor better results appearing earlier in the rankings, $DCG@K$

[92]The true test set accepts submissions at `https://huggingface.co/datasets/ought/raft-submission`.

discounts the *graded_relevance* value of a document according to the rank that it appears at. That is, given that $graded_relevance(d_i)$ is the graded relevance (e.g., 0, 1, 2, or 3, where higher is more relevant) of the ith document:

$$DCG@K = \sum_{i=1}^{K} \frac{graded_relevance(d_i)}{log_2(i+1)} \tag{2}$$

$NDCG@K$ normalizes $DCG@K$ by dividing $DCG@K$ by the $IDCG@K$, which is the $DCG@K$ of ideal ordering of the documents (i.e., sorted according to relevance grades in descending order).[93]

C.1.3 Summarization

ROUGE-2. For summarization, we use the standard `ROUGE` score (Lin, 2004), which consider 2-gram overlap to determine correctness. This is the default accuracy metric for **CNN/DailyMail** and **XSUM**.

C.1.4 Language

BPB. The correctness condition for bits-per-byte (BPB) depends on the average number of bits per byte according to model probabilities; see Gao et al. (2021a). This is the default accuracy metric for **The Pile**, **ICE**, and **TwitterAAE**.

C.1.5 Reasoning

F1 (set match). The correctness condition for F1 (set match) is the F1 score when computed between the sets predicted by the model and the correct reference, which is the default accuracy metric for synthetic reasoning (natural).

Exact match (up to specified indicator). The correctness condition for exact match (up to a specified indicator) is the same as exact match, expect when only looking at the prefix of the prediction and prefix of the reference between the first mention of the specified indicator. This is the default accuracy metric for Dyck and **GSM8K**.

Code. For **HumanEval** (Chen et al., 2021) and **APPS** (Hendrycks et al., 2021c), we use the associated code metrics defined in their respective papers.

Equivalent. The correctness condition for equivalent is if the model generation is mathematically equivalent to the correct reference, which is the default accuracy metric for **MATH**.

Equivalent (chain-of-thought). The correctness condition for equivalent (chain-of-thought) is if the model generation is mathematically equivalent to the correct reference, which is the default accuracy metric for **MATH** (chain-of-thought).

C.2 Calibration and uncertainty

We first setup some formal notation, then state the metrics in the population ("infinite data") setting, and finally give formulas for the metrics that we actually compute on finite data.

Formal setup. We measure calibration metrics for classification tasks. Given an input x, let the true label be $y \in [k] = \{1, \ldots, k\}$ and the model's predicted probability be $p \in [0,1]^k$ where $\sum_j p_j = 1$. Here, p_j denotes the model's confidence that the true label is j. Let $\hat{y} = \arg\max_{j \in [k]} p_j$ be the model's predicted label, and $p^{\max} = \max_{j \in [k]} p_j$ denote the model's confidence in its predicted label $\hat{y}$ (the model's "top probability"). Note that $x, y, p, \hat{y}, p^{\max}$ are all random variables.

[93] More information on these and other ranking metrics can be found at `https://ir-measur.es/en/latest/measures.html`.

Population metrics. We first define the metrics in the population setting—intuitively, the population setting corresponds to having infinite data. The *expected calibration error* (ECE) examines the difference between the model's top probability $p^{\max}$ and the probability the model is correct ($y = \hat{y}$) given $p^{\max}$:

$$\text{ECE} = \mathbb{E}\Big[\big|p^{\max} - \mathbb{E}[y = \hat{y} \mid p^{\max}]\big|\Big] \tag{3}$$

For selective classification, we first need to define two key concepts: coverage and accuracy. Given a threshold $t \geq 0$, the *coverage* $c(t)$ is the fraction of examples for which the model's predicted confidence is at least t:

$$c(t) = P(p^{\max} \geq t). \tag{4}$$

The coverage $c(t)$ is non-increasing in t because when t is higher there will be fewer examples where the model is more confident than t.

The *selective accuracy* $a(t)$ at a threshold $t \geq 0$ is the accuracy for all examples where the model's predicted confidence is at least t:[94]

$$a(t) = \begin{cases} P(y = \hat{y} \mid p^{\max} \geq t) & \text{if } c(t) > 0 \\ 1 & \text{otherwise} \end{cases}. \tag{5}$$

The selective accuracy $a(t)$ is often increasing in t for most machine learning models, however this need not be the case (Jones et al., 2021).

The *selective coverage-accuracy area* (SCAA) plots the selective accuracy (on the y-axis) against the coverage (on the x-axis), and calculates the area under the curve. Formally, we have:

$$\text{SCAA} = \int_{t=0}^{\infty} a(t) dc(t) \tag{6}$$

$$= \int_{t=0}^{\infty} a(t) c'(t) d(t). \tag{7}$$

The *selective-accuracy at 10%* (acc@10%) is the selective accuracy at the threshold t'[95] where the coverage $c(t')$ is 10%—this corresponds to the accuracy on the 10% of examples that the model is most confident on (the model is allowed to abstain on all the remaining examples where it is less confident).

$$\text{acc@10\%} = a(t'), \text{ where } c(t') = 0.1. \tag{8}$$

Finite sample metrics. Suppose we have an evaluation dataset $[(p_1, y_1, \hat{y}_1, p_1^{\max}), \ldots, (p_n, y_n, \hat{y}_n, p_n^{\max})]$, where each $(p_i, y_i, \hat{y}_i, p_i^{\max})$ is sampled independently from the same distribution as $(p, y, \hat{y}, p^{\max})$ defined in the Setup above. Recall that p is the model's predicted probability vector, y is the true label, $\hat{y}$ is the model's predicted label, and $p^{\max} = p_{\hat{y}}$ is the model's probability for its prediction $\hat{y}$ (the model's "top probability"). For convenience, suppose that the evaluation dataset is sorted in decreasing order of the model's top-probability $p^{\max}$, that is for all $i \leq j$, we have $p_i^{\max} \geq p_j^{\max}$.

Following prior work, we evaluate the ECE by binning the model's predictions into m bins. Note that some form of binning is necessary, otherwise we cannot estimate the conditional expectation $E[y = \hat{y} \mid p^{\max}]$. We select a set of m intervals ("bins") $I_1, \ldots I_k$ that partition $[0, 1]$ (that is, the intervals do not overlap and they cover $[0, 1]$). We choose uniform-mass bins (Kumar et al., 2019), where an equal number of $p^{\max}$ values fall into each bin.

Formally, we define the bin-count n_i of a bin i as:

$$n_i = \sum_{j=1}^{n} \mathbb{I}(p_j^{\max} \in I_i) \tag{9}$$

[94] We have a special case in the definition when $c(t) = P(p^{\max} \geq t) = 0$. This is because the definition of the selective accuracy involves a conditional expectation, which would be undefined if the condition never holds. However, this is just for mathematical convenience and the reader should feel free to ignore the edge case for simplicity.

[95] There can be multiple values t where $c(t) = 0$, however this will in fact not change the definition because if $c(t_1) = c(t_2)$ where $t_1 \leq t_2$ then $P(t_1 \leq p^{\max} \leq t_2) = 0$, because $c(t_2) = c(t_1) + P(t_1 \leq p^{\max} \leq t_2) = 0$ and all these quantities are non-negative. The reader should feel free to ignore this edge case for simplicity.